Tutorial: exam-style practice

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Part A: 12 multiple-choice questions (1 mark each). Part B: 5 structured questions (36 marks). Same format as the final exam.

Answer Part A by circling one option. Answer Part B in full, showing your working. The answer key is at the back.

Part A · Multiple choice

Choose one answer for each question.

A1 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO1

Which alcohol is tertiary?

A. 2-methyl-2-propanol
B. 2-propanol
C. 1-butanol
D. 2-methyl-1-propanol

A2 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO1

Arrange in order of increasing boiling point: ethanol, methanol, 1-propanol. (Ball 14.E)

A. methanol < ethanol < 1-propanol
B. 1-propanol < ethanol < methanol
C. ethanol < methanol < 1-propanol
D. all equal

A3 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO2

In an SN2 reaction, the nucleophile attacks:

A. from the side opposite the leaving group
B. a carbocation
C. the leaving group
D. the hydrogen atoms

A4 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO2

Which statement about SN1 is correct?

A. Its rate depends only on [haloalkane]
B. It occurs in one step
C. It is fastest for 1° haloalkanes
D. It inverts a chiral carbon completely

A5 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO3

Oxidation of 2-propanol gives:

A. propanone
B. propanal
C. propanoic acid
D. propene

A6 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO3

Ethanol with conc. H_2SO_4 at 140 °C and excess ethanol gives mainly:

A. diethyl ether
B. ethene
C. ethanal
D. ethanoic acid

A7 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO4

Ethers have lower boiling points than isomeric alcohols because they:

A. cannot hydrogen bond to each other
B. are ionic
C. are heavier
D. contain sulfur

A8 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO4

The functional group of a thiol is:

A. –SH
B. –S–S–
C. –OH
D. $-SO_4$

A9 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO5

The IUPAC name of CH_3CH_2CHO is:

A. propanal
B. propanone
C. propanol
D. 1-propanal

A10 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO5

Which reagent gives a silver mirror with aldehydes?

A. Tollens'
B. Benedict's
C. bromine water
D. conc. H_2SO_4

A11 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO5

Which alcohol is oxidised to 3-pentanone?

A. 3-pentanol
B. 1-pentanol
C. 2-methyl-2-butanol
D. pentanal

A12 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO5

Formalin, used to preserve specimens, is a solution of:

A. methanal (formaldehyde)
B. ethanal
C. propanone
D. methanol

Part B · Structured questions

Write your answers on paper, then self-mark with the worked solution.

Question B1 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO1 6 marks, C3

(a) Name and classify (1°, 2° or 3°) each alcohol: (i) $CH_3CH_2CH(OH)CH_3$ (ii) $(CH_3)_2CHCH_2OH$ (iii) $(CH_3)_2C(OH)CH_2CH_3$. (3 marks) (b) Methanol (32 g/mol) boils at 65 °C but ethane (30 g/mol) boils at −89 °C. Explain. (2 marks) (c) Explain why 1-octanol is much less soluble in water than ethanol. (1 mark)

Question B2 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO2 9 marks, C4

Two haloalkanes are converted into alcohols: **P** = 1-bromobutane with warm aqueous NaOH; **Q** = 2-bromo-2-methylpropane with water. (a) State the mechanism (SN1 or SN2) for each and give one reason for each choice. (4 marks) (b) Draw the structure of the alcohol formed in each case and name it. (2 marks) (c) For **Q**, draw the intermediate and name its type. (2 marks) (d) What would you expect if 1-bromobutane were heated with KOH in ethanol instead? (1 mark)

Question B3 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO3 7 marks, C3

Three unlabelled bottles contain 1-propanol, 2-propanol and 2-methyl-2-propanol. (a) Describe what you would see when each is warmed with acidified potassium dichromate, and name any organic product. (5 marks) (b) Write an equation for the dehydration of 2-propanol to an alkene, stating the conditions. (2 marks)

Question B4 Topic: Organic Compounds that Contain Oxygen and Sulphur | CH3-LO4 | 6 marks, C2

(a) Draw and give the common name of the ether $CH_3CH_2OCH(CH_3)_2$. (2 marks) (b) Explain why diethyl ether was replaced as an anaesthetic. (1 mark) (c) Write an equation for the oxidation of ethanethiol to diethyl disulfide. (2 marks) (d) State why disulfide bridges are important in proteins. (1 mark)

Question B5 Topic: Organic Compounds that Contain Oxygen and Sulphur | CH3-LO5 | 8 marks, C4

Compounds **X** and **Y** both have the formula C_3H_6O. X gives a silver mirror with Tollens' reagent; Y does not. (a) Identify X and Y, giving their structures and IUPAC names. (2 marks) (b) Explain the Tollens' results. (2 marks) (c) Name the product when X is oxidised. (1 mark) (d) Give the products when X and Y are reduced with H_2/Ni, and classify each alcohol. (2 marks) (e) Explain why propanal (bp 49 °C) boils higher than butane (bp −1 °C). (1 mark)

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Adapted from *The Basics of GOB Chemistry* (Ball, Hill & Scott) chapter 14 and exercises 14.E, and *Fundamentals of General Organic and Biological Chemistry* 14.4–14.9 and 15.1–15.5, LibreTexts, CC BY-NC-SA 3.0. SN1 and SN2 items, colour tests and pharmacy scenarios are original material by Dr. NAM. Shared under CC BY-NC-SA for non-commercial teaching.

PHD125 Pharmaceutical Organic Chemistry · Organic Compounds that Contain Oxygen and Sulphur

Tutorial: exam-style practice

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Part A: 12 multiple-choice questions (1 mark each). Part B: 5 structured questions (36 marks). Same format as the final exam.

Answer key

Part A · Multiple choice

A1 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO1

Which alcohol is tertiary?

A. 2-methyl-2-propanol

B. 2-propanol

C. 1-butanol

D. 2-methyl-1-propanol

The C–OH carbon is bonded to three carbons.

A2 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO1

Arrange in order of increasing boiling point: ethanol, methanol, 1-propanol. (Ball 14.E)

A. methanol < ethanol < 1-propanol

B. 1-propanol < ethanol < methanol

C. ethanol < methanol < 1-propanol

D. all equal

Larger molecules, stronger dispersion forces in addition to H-bonding.

A3 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO2

In an SN2 reaction, the nucleophile attacks:

A. from the side opposite the leaving group

B. a carbocation

C. the leaving group

D. the hydrogen atoms

Backside attack, one step.

A4 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO2

Which statement about SN1 is correct?

A. Its rate depends only on [haloalkane]

B. It occurs in one step

C. It is fastest for 1° haloalkanes

D. It inverts a chiral carbon completely

The slow step is formation of the carbocation.

A5 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO3

Oxidation of 2-propanol gives:

A. propanone

B. propanal

C. propanoic acid

D. propene

2° → ketone.

A6 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO3

Ethanol with conc. H_2SO_4 at 140 °C and excess ethanol gives mainly:

A. diethyl ether

B. ethene

C. ethanal

D. ethanoic acid

Intermolecular dehydration (Ball 14.E).

A7 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO4

Ethers have lower boiling points than isomeric alcohols because they:

A. cannot hydrogen bond to each other

B. are ionic

C. are heavier

D. contain sulfur

No O–H in ethers.

A8 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO4

The functional group of a thiol is:

A. –SH

B. –S–S–

C. –OH

D. $-SO_4$

R–SH (mercaptans).

A9 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO5

The IUPAC name of CH_3CH_2CHO is:

A. propanal

B. propanone

C. propanol

D. 1-propanal

Aldehyde: -al; C1 needs no number.

A10 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO5

Which reagent gives a silver mirror with aldehydes?

A. Tollens'

B. Benedict's

C. bromine water

D. conc. H_2SO_4

Ag^+ is reduced to Ag.

A11 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO5

Which alcohol is oxidised to 3-pentanone?

A. 3-pentanol

B. 1-pentanol

C. 2-methyl-2-butanol

D. pentanal

2° alcohol at C3.

A12 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO5

Formalin, used to preserve specimens, is a solution of:

A. methanal (formaldehyde)

B. ethanal

C. propanone

D. methanol

37–40% formaldehyde in water (FGOB 15.4).

Part B · Structured questions

Question B1 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO1 6 marks, C3

(a) Name and classify (1°, 2° or 3°) each alcohol: (i) $CH_3CH_2CH(OH)CH_3$ (ii) $(CH_3)_2CHCH_2OH$ (iii) $(CH_3)_2C(OH)CH_2CH_3$. (3 marks) (b) Methanol (32 g/mol) boils at 65 °C but ethane (30 g/mol) boils at −89 °C. Explain. (2 marks) (c) Explain why 1-octanol is much less soluble in water than ethanol. (1 mark)

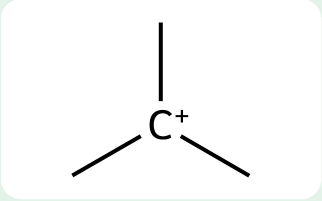
(a)(i) 2-butanol, 2° (ii) 2-methyl-1-propanol, 1° (iii) 2-methyl-2-butanol, 3°. (b) Methanol molecules form hydrogen bonds through their –OH groups, which need much more energy to break than the weak dispersion forces between nonpolar ethane molecules. (c) The long nonpolar C_8 chain dominates, so 1-octanol behaves more like a hydrocarbon; ethanol's single OH is enough for its short chain.

Step	Marking criteria	Marks
1	(i) 2-butanol, 2°	1
2	(ii) 2-methyl-1-propanol, 1°	1
3	(iii) 2-methyl-2-butanol, 3°	1
4	Methanol: hydrogen bonding	1
5	Ethane: only weak dispersion forces	1
6	Long nonpolar chain in 1-octanol	1
	Total	**6**

Question B2 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO2 9 marks, C4

Two haloalkanes are converted into alcohols: **P** = 1-bromobutane with warm aqueous NaOH; **Q** = 2-bromo-2-methylpropane with water. (a) State the mechanism (SN1 or SN2) for each and give one reason for each choice. (4 marks) (b) Draw the structure of the alcohol formed in each case and name it. (2 marks) (c) For **Q**, draw the intermediate and name its type. (2 marks) (d) What would you expect if 1-bromobutane were heated with KOH in ethanol instead? (1 mark)

(a) P: SN2: it is a primary haloalkane (uncrowded for backside attack) with a strong nucleophile (OH^-). Q: SN1: it is tertiary (too crowded for SN2) and forms a stable 3° carbocation; water is a weak nucleophile. (b) P → $CH_3CH_2CH_2CH_2OH$, 1-butanol. Q → $(CH_3)_3COH$, 2-methyl-2-propanol.



(c) $(CH_3)_3C^+$, a tertiary carbocation (drawn above). (d) Elimination: but-1-ene, $CH_2{=}CHCH_2CH_3$, forms instead of the alcohol.

Step	Marking criteria	Marks
1	P: SN2	1
2	Reason for P (1° or strong nucleophile)	1
3	Q: SN1	1
4	Reason for Q (3°, stable carbocation)	1
5	1-butanol	1
6	2-methyl-2-propanol	1
7	Carbocation drawn correctly	1
8	Named tertiary carbocation	1
9	Elimination to an alkene (1-butene)	1
	Total	**9**

Question B3 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO3 7 marks, C3

Three unlabelled bottles contain 1-propanol, 2-propanol and 2-methyl-2-propanol. (a) Describe what you would see when each is warmed with acidified potassium dichromate, and name any organic product. (5 marks) (b) Write an equation for the dehydration of 2-propanol to an alkene, stating the conditions. (2 marks)

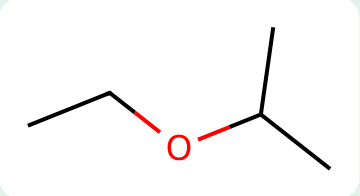
(a) 1-Propanol: orange → green; product propanal (then propanoic acid with excess oxidant). 2-Propanol: orange → green; product propanone. 2-Methyl-2-propanol: stays orange, no reaction (3°). (b) $CH_3CH(OH)CH_3 \rightarrow CH_3CH{=}CH_2 + H_2O$, with excess concentrated H_2SO_4 at about 180 °C.

Step	Marking criteria	Marks
1	Orange → green for 1° and 2°	1
2	1-propanol → propanal (or propanoic acid)	1
3	2-propanol → propanone	1
4	3° stays orange	1
5	Reason: no H on the C–OH carbon	1
6	Correct equation to propene	1
7	Conditions: conc. H_2SO_4, ~180 °C	1
	Total	**7**

Question B4 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO4 6 marks, C2

(a) Draw and give the common name of the ether $CH_3CH_2OCH(CH_3)_2$. (2 marks) (b) Explain why diethyl ether was replaced as an anaesthetic. (1 mark) (c) Write an equation for the oxidation of ethanethiol to diethyl disulfide. (2 marks) (d) State why disulfide bridges are important in proteins. (1 mark)

(a) Ethyl isopropyl ether (Ball 14.E).



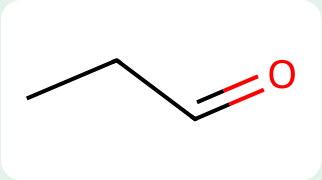
(b) It is highly flammable and causes nausea; halogenated ethers are safer to use. (c) 2 CH_3CH_2SH + [O] → $CH_3CH_2S–SCH_2CH_3$ + H_2O. (d) They link parts of protein chains (between cysteine residues), holding the protein in its shape.

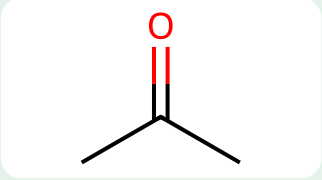
Step	Marking criteria	Marks
1	Correct structure	1
2	Ethyl isopropyl ether	1
3	Flammable / nausea	1
4	Correct reactants and product	1
5	Balanced with H_2O and [O]	1
6	Holds protein shape (cysteine links)	1
	Total	**6**

Question B5 Topic: Organic Compounds that Contain Oxygen and Sulphur CH3-LO5 8 marks, C4

Compounds **X** and **Y** both have the formula C_3H_6O. X gives a silver mirror with Tollens' reagent; Y does not. (a) Identify X and Y, giving their structures and IUPAC names. (2 marks) (b) Explain the Tollens' results. (2 marks) (c) Name the product when X is oxidised. (1 mark) (d) Give the products when X and Y are reduced with H_2/Ni, and classify each alcohol. (2 marks) (e) Explain why propanal (bp 49 °C) boils higher than butane (bp −1 °C). (1 mark)

(a) X = propanal, CH_3CH_2CHO; Y = propanone, CH_3COCH_3.





(b) The aldehyde has an H on its carbonyl carbon, so it is easily oxidised; it reduces Ag^+ to Ag (silver mirror). The ketone resists oxidation. (c) Propanoic acid. (d) X → 1-propanol (1°); Y → 2-propanol (2°). (e) The polar C=O gives strong dipole–dipole forces; butane has only dispersion forces.

Step	Marking criteria	Marks
1	X propanal (structure)	1
2	Y propanone (structure)	1
3	Aldehyde easily oxidised (H on C=O carbon)	1
4	Ag^+ reduced to Ag; ketone does not react	1
5	Propanoic acid	1
6	1-propanol, 1°	1
7	2-propanol, 2°	1
8	Dipole–dipole vs dispersion	1
	Total	**8**

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